

# Precalculus Toolbox

This appendix collects the precalculus tools you will want on hand but that many precalculus courses now treat briefly or omit: the reciprocal trigonometric functions (§A.1), the product-to-sum and sum-to-product identities (§A.2), sigma notation, power sums, and the geometric series (§A.3), partial fractions (§A.4), a quick tour of the conic sections (§A.5), and absolute value and inequalities (§A.6). Some are pointed to from the main text at the moment they are first needed; others are here as a handy reference to reach for when you want them. The sections are independent of one another; read each when the main text sends you, or straight through as a warm-up. Nothing here uses calculus, with one small exception flagged in §A.3, where an infinite sum is read as a limit.

**By the end of this appendix, you will be able to:**

- evaluate the reciprocal trigonometric functions  $\sec$ ,  $\csc$ , and  $\cot$ , and apply their Pythagorean identities with the correct signs;
- derive every product-to-sum and sum-to-product identity from the angle-sum formulas — and recover any one you forget;
- manipulate sums in sigma notation, evaluate the power sums  $\sum k$ ,  $\sum k^2$ , and  $\sum k^3$ , and sum a geometric series, finite or infinite;
- split a proper rational function into partial fractions;
- read off each conic section's key features — centre, vertices, foci, and axes — from its standard equation;
- unpack an absolute-value inequality  $|x - a| < \delta$  into an interval, and apply the triangle inequality.

---

## A.1 The Reciprocal Trigonometric Functions

The three primary trigonometric functions  $\sin$ ,  $\cos$ , and  $\tan$  have three companions — two honest reciprocals and one inverted ratio. They carry no new information — every fact about them is a fact about  $\sin$  and  $\cos$  in disguise. Yet calculus speaks them fluently: the derivative of  $\tan x$  turns out to be  $\sec^2 x$ , the inverse secant needs a principal range of its own, and integral tables are full of secants and cosecants. This section gives them an honest introduction.

**DEFINITION A.1**

For any angle  $\theta$  at which the denominator is nonzero, the *secant*, *cosecant*, and *cotangent* functions are defined by

$$\sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

*Informally: secant is the reciprocal of cosine, cosecant is the reciprocal of sine, and cotangent flips the ratio that defines tangent. The names cross over historically — secant with cosine, cosecant with sine — and the pairing simply has to be memorized.*

In a right triangle, the definitions invert the familiar ratios. Since  $\cos \theta$  is adjacent over hypotenuse,  $\sec \theta$  is hypotenuse over adjacent; since  $\sin \theta$  is opposite over hypotenuse,  $\csc \theta$  is hypotenuse over opposite; and  $\cot \theta$  is adjacent over opposite. Special-angle values follow instantly from the primary ones: for instance  $\sec \frac{\pi}{3} = 1/\cos \frac{\pi}{3} = 2$ , and  $\cot \frac{\pi}{4} = 1$ . Each function is undefined exactly where its denominator vanishes — sec and tan at odd multiples of  $\frac{\pi}{2}$ , csc and cot at multiples of  $\pi$  — and near those angles its values blow up, producing the vertical asymptotes on their graphs.

**CAUTION**

Cotangent is defined as  $\cos \theta / \sin \theta$ , *not* as  $1/\tan \theta$ . On most of the domain the two agree, but not everywhere: at  $\theta = \frac{\pi}{2}$  we have  $\cot \frac{\pi}{2} = \cos \frac{\pi}{2} / \sin \frac{\pi}{2} = 0/1 = 0$ , a perfectly good value, while  $1/\tan \frac{\pi}{2}$  is meaningless —  $\tan \frac{\pi}{2}$  is undefined. The quotient  $\cos / \sin$  is the robust definition;  $1/\tan$  is a shortcut valid only where  $\tan \theta$  is defined and nonzero.

Dividing the Pythagorean identity  $\sin^2 \theta + \cos^2 \theta = 1$  by  $\cos^2 \theta$  or by  $\sin^2 \theta$  produces the two companion identities that calculus uses constantly.

**PROPOSITION A.1** *Pythagorean companions*

$$\begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta & (\cos \theta \neq 0), \\ 1 + \cot^2 \theta &= \csc^2 \theta & (\sin \theta \neq 0). \end{aligned}$$

**PROOF**

Divide  $\sin^2 \theta + \cos^2 \theta = 1$  through by  $\cos^2 \theta$ :

$$\frac{\sin^2 \theta}{\cos^2 \theta} + 1 = \frac{1}{\cos^2 \theta}, \quad \text{that is,} \quad \tan^2 \theta + 1 = \sec^2 \theta.$$

Dividing instead by  $\sin^2 \theta$  gives  $1 + \cot^2 \theta = \csc^2 \theta$ . □

**EXAMPLE A.1**

Evaluate the following.

1.  $\sec\left(\frac{2\pi}{3}\right)$  and  $\cot\left(-\frac{\pi}{4}\right)$ .
2. Given that  $\tan \theta = 3$  and  $\theta$  lies in the third quadrant, find  $\sec \theta$  and  $\csc \theta$ .

**SOLUTION**

1. Since  $\cos \frac{2\pi}{3} = -\frac{1}{2}$ , we get  $\sec \frac{2\pi}{3} = 1/(-\frac{1}{2}) = -2$ . Since  $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$  and  $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$ , we get  $\cot(-\frac{\pi}{4}) = \frac{\sqrt{2}/2}{-\sqrt{2}/2} = -1$ .
2. By Proposition A.1,  $\sec^2 \theta = 1 + \tan^2 \theta = 1 + 9 = 10$ , so  $\sec \theta = \pm\sqrt{10}$ . In the third quadrant  $\cos \theta < 0$ , hence  $\sec \theta < 0$ , and therefore  $\sec \theta = -\sqrt{10}$ . For the cosecant,  $\cot \theta = 1/\tan \theta = \frac{1}{3}$  (legitimate here, since  $\tan \theta = 3$  is defined and nonzero), so  $\csc^2 \theta = 1 + \cot^2 \theta = 1 + \frac{1}{9} = \frac{10}{9}$  and  $\csc \theta = \pm\frac{\sqrt{10}}{3}$ . In the third quadrant  $\sin \theta < 0$ , so  $\csc \theta = -\frac{\sqrt{10}}{3}$ . The squaring in Proposition A.1 forgets all signs; the quadrant restores them.

That is the whole toolkit: three definitions, the right-triangle readings, and the two Pythagorean companions — every fact about  $\sec$ ,  $\csc$ , and  $\cot$  reduced to a fact about  $\sin$  and  $\cos$ .

## A.2 The Product-to-Sum and Sum-to-Product Identities

One algebraic step in this book carries more weight than any other: rewriting the difference  $\sin(x+h) - \sin x$  as a *product*, which is what makes the derivative of the sine function computable. The identity that does it belongs to a family — the product-to-sum and sum-to-product identities — that older curricula drilled and most current precalculus courses omit. The good news is that the whole family is nothing but the angle-sum formulas, added and subtracted in pairs. If you learn one thing from this section, learn how to *rebuild* the identities; the memorizing then becomes optional.

Everything starts from the four angle-sum and angle-difference formulas, which we assume and record here for reference:

$$\begin{aligned}\sin(u+v) &= \sin u \cos v + \cos u \sin v, \\ \sin(u-v) &= \sin u \cos v - \cos u \sin v, \\ \cos(u+v) &= \cos u \cos v - \sin u \sin v, \\ \cos(u-v) &= \cos u \cos v + \sin u \sin v.\end{aligned}$$

Add or subtract these in matching pairs and half of each right-hand side cancels, leaving a single product.

**PROPOSITION A.2** *Product-to-sum identities*

$$\begin{aligned}\sin u \cos v &= \frac{1}{2}[\sin(u+v) + \sin(u-v)], \\ \cos u \cos v &= \frac{1}{2}[\cos(u+v) + \cos(u-v)], \\ \sin u \sin v &= \frac{1}{2}[\cos(u-v) - \cos(u+v)].\end{aligned}$$

**PROOF**

Add the two sine formulas: the  $\cos u \sin v$  terms cancel, leaving  $\sin(u+v) + \sin(u-v) = 2 \sin u \cos v$ ; divide by 2. Add the two cosine formulas: the  $\sin u \sin v$  terms cancel, giving  $\cos(u+v) + \cos(u-v) = 2 \cos u \cos v$ . Subtract the cosine formulas instead —  $\cos(u-v) - \cos(u+v)$  — and the  $\cos u \cos v$  terms cancel, giving  $2 \sin u \sin v$ .  $\square$

Setting  $u = v = \theta$  in these product-to-sum identities pays an immediate dividend. The middle line becomes  $\cos^2 \theta = \frac{1}{2}[\cos 2\theta + \cos 0]$  and the last becomes  $\sin^2 \theta = \frac{1}{2}[\cos 0 - \cos 2\theta]$ ; since  $\cos 0 = 1$ , these are the *power-reduction* formulas

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2},$$

which trade a *squared* trigonometric function for a plain one at twice the angle. (The first line,  $\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$ , is the double-angle formula for sine read the same way.) These are the forms integration reaches for to handle  $\sin^2$  and  $\cos^2$  when the techniques of integration arrive.

The sum-to-product identities are the same cancellations read in the other direction. The only new idea is a change of names: given a sum like  $\sin A + \sin B$ , arrange for the two angles to read  $u+v$  and  $u-v$  by choosing  $u$  and  $v$  so that  $u+v = A$  and  $u-v = B$ . Adding these two equations gives  $2u = A+B$  and subtracting them gives  $2v = A-B$ , so  $u$  is the *average* and  $v$  the *half-difference*:

$$u = \frac{A+B}{2}, \quad v = \frac{A-B}{2}.$$

**PROPOSITION A.3** *Sum-to-product identities*

$$\begin{aligned}\sin A + \sin B &= 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}, \\ \sin A - \sin B &= 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}, \\ \cos A + \cos B &= 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}, \\ \cos A - \cos B &= -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}.\end{aligned}$$

**PROOF**

With  $u = \frac{A+B}{2}$  and  $v = \frac{A-B}{2}$  we have  $u+v = A$  and  $u-v = B$ . Then, reading the cancellations in the previous proof from right to left,

$$\begin{aligned}\sin A + \sin B &= \sin(u+v) + \sin(u-v) \\ &= 2 \sin u \cos v = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2},\end{aligned}$$

and the other three lines follow the same way:  $\sin A - \sin B = \sin(u+v) - \sin(u-v) = 2 \cos u \sin v$  (this time subtracting the two sine expansions);  $\cos A + \cos B = 2 \cos u \cos v$ ; and  $\cos A - \cos B = \cos(u+v) - \cos(u-v) = -2 \sin u \sin v$ . Note the minus sign in the last line — the only one in the family, and the easiest thing to misremember; the subtraction order  $\cos(u+v) - \cos(u-v)$  is where it comes from.  $\square$

**STRATEGY A.1** *Recovering an identity you forgot*

1. Write down the two angle-sum (or angle-difference) expansions that contain the pieces you have, and add or subtract them so that the unwanted terms cancel.
2. If your expression is a sum of two trig values at angles  $A$  and  $B$ , substitute the average  $u = \frac{A+B}{2}$  and half-difference  $v = \frac{A-B}{2}$ , so that  $A = u + v$  and  $B = u - v$ .

*Rebuilding takes perhaps a minute; a memorized list of eight is easy to get wrong by a sign. Verifying is even cheaper — expand the product side with the angle-sum formulas and watch the cross terms cancel.*

**EXAMPLE A.2**

Rewrite each expression as requested.

1. Write  $\sin(x+h) - \sin x$  as a product.
2. Write  $\sin 5x \sin 3x$  as a sum.

**SOLUTION**

1. Apply the sum-to-product identity for  $\sin A - \sin B$  with  $A = x + h$  and  $B = x$ . The average is  $\frac{A+B}{2} = x + \frac{h}{2}$  and the half-difference is  $\frac{A-B}{2} = \frac{h}{2}$ , so

$$\sin(x+h) - \sin x = 2 \cos\left(x + \frac{h}{2}\right) \sin \frac{h}{2}.$$

This is precisely the rewriting the derivative of sine turns on: the difference of two sines becomes a product with the small quantity  $h/2$  isolated inside a single sine.

2. Apply the third product-to-sum identity with  $u = 5x$  and  $v = 3x$ :

$$\begin{aligned} \sin 5x \sin 3x &= \frac{1}{2} [\cos(5x - 3x) - \cos(5x + 3x)] \\ &= \frac{1}{2} [\cos 2x - \cos 8x]. \end{aligned}$$

Turning products into sums is the direction integration will want: a sum of cosines is far easier to integrate than a product.

Of the whole family, the identity for  $\sin A - \sin B$  in Proposition A.3 is the one worth keeping at your fingertips; the rest follow from the same two moves the moment you need them.

## A.3 Sigma Notation, Power Sums, and the Geometric Series

Before the geometric series itself, a word on the notation that names it and a few moves for handling sums cleanly. The capital-sigma symbol  $\sum$  abbreviates a sum:  $\sum_{k=1}^n a_k$  means  $a_1 + a_2 + \cdots + a_n$ , the terms  $a_k$  added as the *index*  $k$  climbs through the whole numbers from the lower limit to the upper. The index is only a placeholder —  $\sum_{k=1}^n a_k$  and  $\sum_{j=1}^n a_j$  denote the very same sum — and an upper limit of  $\infty$ , as in  $\sum_{k=0}^{\infty} a_k$ , signals a sum that never stops, to be read as the limit of its running totals in the sense settled just below.

Three moves are worth having at hand. *Re-indexing* renames the counter without disturbing the sum: setting  $j = k - 1$  rewrites  $\sum_{k=1}^n a_k$  as  $\sum_{j=0}^{n-1} a_{j+1}$ , the same terms relabelled. *Peeling* detaches the first or last term, as in  $\sum_{k=0}^n a_k = a_0 + \sum_{k=1}^n a_k$ . And *splitting at a cutoff*  $n_0$  breaks a sum into a leading part and a *tail*,

$$\sum_{k=0}^n a_k = \sum_{k=0}^{n_0} a_k + \sum_{k=n_0+1}^n a_k,$$

a standard way to control a series: bound its tail while the leading partial sum is set aside.

One special sum collapses outright. When each term is the *difference* of consecutive values of a sequence  $b_k$ , the interior pieces cancel in pairs:

$$\sum_{k=0}^m (b_k - b_{k+1}) = b_0 - b_{m+1},$$

since every  $b_k$  with  $1 \leq k \leq m$  is added once and subtracted once. A sum that collapses this way is said to *telescope*, after the way a spyglass folds shut — and it is the exact mechanism behind both the geometric-sum formula and the power sums below.

Those moves are all finite bookkeeping. Adding up *infinitely* many terms is a genuinely new act, and most of its theory belongs to a later chapter of calculus. But one infinite sum is so simple, and so useful so early, that it deserves to be settled now: the geometric series, in which each term is a fixed multiple  $r$  of the one before. It earns its keep everywhere — the *tail* of a series (its leftover terms past a cutoff) can often be trapped under a geometric tail and squeezed to nothing. Everything rests on one finite identity plus one limit.

### PROPOSITION A.4 *Finite geometric sum*

For any number  $r \neq 1$  and any positive integer  $m$ ,

$$1 + r + r^2 + \cdots + r^m = \frac{1 - r^{m+1}}{1 - r}.$$

**PROOF**

Write  $S = 1 + r + r^2 + \cdots + r^m$  and multiply by  $1 - r$ :

$$(1 - r)S = (1 + r + \cdots + r^m) - (r + r^2 + \cdots + r^{m+1}) = 1 - r^{m+1},$$

because each power  $r, r^2, \dots, r^m$  appears once in each bracket and cancels, leaving only the leading 1 and the final  $-r^{m+1}$  — the sum telescopes. Dividing by  $1 - r$  (legitimate, since  $r \neq 1$ ) gives the identity.  $\square$

Now let the sum run on. If  $|r| < 1$ , each multiplication by  $r$  shrinks the next term by a fixed factor, and the power  $r^{m+1}$  in the formula dies away: its size  $|r|^{m+1}$  drops below any positive threshold once  $m$  is large enough. In the arrow notation for a limit (here the whole-number index  $m$  runs off to infinity, rather than a variable approaching a finite point),  $r^{m+1} \rightarrow 0$  as  $m \rightarrow \infty$ , so the running totals

$$S_m = \frac{1 - r^{m+1}}{1 - r} \quad \text{settle toward} \quad \frac{1 - 0}{1 - r} = \frac{1}{1 - r}.$$

The value of an infinite sum *means* exactly this: the limit of its running totals. (The running totals also go by the name *partial sums*; the completeness facts behind general convergence arguments belong to a later chapter of calculus, and none of them are needed for the simple limit here.)

**PROPOSITION A.5** *Infinite geometric series*

If  $|r| < 1$ , then for any number  $a$ ,

$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots = \frac{a}{1 - r},$$

and in particular the tail starting at  $k = 1$  sums to

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1 - r}.$$

**PROOF**

By Proposition A.4 the  $m$ -th running total is  $a \cdot \frac{1 - r^{m+1}}{1 - r}$ . As  $m \rightarrow \infty$  the power  $r^{m+1}$  tends to 0, so the totals tend to  $\frac{a}{1 - r}$ . The tail formula is the case  $a = r$ :  $r + r^2 + r^3 + \cdots = \frac{r}{1 - r}$ .  $\square$

**EXAMPLE A.3**

Evaluate each infinite sum.

1.  $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$
2.  $0.999 \dots$ , read as  $\frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \cdots$

**SOLUTION**

1. This is Proposition A.5 with  $a = \frac{1}{2}$  and  $r = \frac{1}{2}$ :

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1/2}{1 - 1/2} = 1.$$

The picture matches the algebra: eat half the cake, then half of what remains, then half of that — the running total approaches one whole cake, and in the limit exactly fills it.

2. Here  $a = \frac{9}{10}$  and  $r = \frac{1}{10}$ :

$$0.999 \dots = \frac{9/10}{1 - 1/10} = \frac{9/10}{9/10} = 1.$$

Not approximately 1 — exactly 1. The two decimal expressions 0.999... and 1 are two names for the same number, and the infinite-sum reading is what makes that a computation rather than a debate.

**CAUTION**

The hypothesis  $|r| < 1$  is doing real work. For  $r = 2$  the series  $1 + 2 + 4 + 8 + \cdots$  grows without bound and has no sum, while the formula, applied blindly, would report  $\frac{1}{1-2} = -1$  — nonsense. The formula  $\frac{a}{1-r}$  does not *create* a sum; it reports the limit of the running totals, and — unless every term is zero — that limit exists only when  $|r| < 1$ .

The geometric series is one of two finite-sum patterns worth carrying forward. The other is a short family of *power sums* — the totals of the first  $n$  whole numbers, their squares, and their cubes — which surface the moment Chapter 6 builds the integral as a limit of sums over  $n$  equal pieces. Like the geometric sum, they drop straight out of telescoping.

**PROPOSITION A.6** *Power sums*

For every positive integer  $n$ ,

$$\begin{aligned}\sum_{k=1}^n k &= \frac{n(n+1)}{2}, \\ \sum_{k=1}^n k^2 &= \frac{n(n+1)(2n+1)}{6}, \\ \sum_{k=1}^n k^3 &= \left[ \frac{n(n+1)}{2} \right]^2.\end{aligned}$$

**PROOF**

Each formula comes from the same collapse that gave the geometric sum.



Telescoping supplies  $\sum_{k=1}^n [(k+1)^2 - k^2] = (n+1)^2 - 1$ , since every interior square is added once and subtracted once; and expanding the summand,  $(k+1)^2 - k^2 = 2k + 1$ . Equating the two evaluations,

$$2 \sum_{k=1}^n k + n = (n+1)^2 - 1 = n^2 + 2n,$$

which solves to  $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ . Climbing one power,  $\sum_{k=1}^n [(k+1)^3 - k^3] = (n+1)^3 - 1$  with  $(k+1)^3 - k^3 = 3k^2 + 3k + 1$  gives

$$3 \sum_{k=1}^n k^2 + 3 \sum_{k=1}^n k + n = (n+1)^3 - 1;$$

substituting the value of  $\sum k$  just found and solving leaves  $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ . The cube sum follows by the identical step one power higher: from  $(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1$ , telescoping gives  $4 \sum_{k=1}^n k^3 + 6 \sum k^2 + 4 \sum k + n = (n+1)^4 - 1$ , and substituting the two sums already found leaves  $4 \sum k^3 = n^2(n+1)^2$ , that is  $\sum_{k=1}^n k^3 = [\frac{n(n+1)}{2}]^2$ .  $\square$

A quick check pins the shapes down: at  $n = 3$  the square sum reads  $1+4+9 = 14$ , and  $\frac{3 \cdot 4 \cdot 7}{6} = 14$  agrees. What matters is less any single value than the whole shape at once — to add the areas of  $n$  thin rectangles under  $y = x^2$  and watch the total settle as  $n \rightarrow \infty$ , one needs  $\sum k^2$  as a formula in  $n$ , not a number. That is exactly the step behind the first integral computed from its definition in Chapter 6.

Two formulas from this section will see the most reuse: the geometric tail bound  $\sum_{k \geq 1} r^k = \frac{r}{1-r}$  of Proposition A.5, behind any argument that traps a series under a geometric tail; and the square sum  $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$  of Proposition A.6, behind the first integrals Chapter 6 computes from scratch.

## A.4 Partial Fractions

Adding two fractions over a common denominator is routine; the reverse — breaking one complicated fraction back into simple pieces — is *partial fractions*. A rational function is a quotient of polynomials  $P(x)/Q(x)$ , and partial fractions rewrites it as a sum of fragments simple enough to handle one at a time. The smallest instance is worth seeing first:

$$\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1},$$

as recombining the right-hand side over a common denominator confirms. (Read left to right, this is exactly the split that makes  $\sum \frac{1}{k(k+1)}$  telescope — §A.3's collapsing sum in disguise.) Its real payoff is in integration, where each fragment turns out to have a standard antiderivative.

### PROPOSITION A.7 Partial-fraction decomposition

Let  $P(x)/Q(x)$  be a proper rational function — one whose numerator has lower

degree than its denominator — and factor  $Q$  completely into linear and irreducible quadratic factors. Then  $P/Q$  equals a sum of partial fractions, contributed factor by factor:

- each distinct linear factor  $(x - a)$  contributes one term  $\frac{A}{x - a}$ ;
- each repeated linear factor  $(x - a)^m$  contributes  $\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_m}{(x - a)^m}$ ;
- each irreducible quadratic factor  $(x^2 + px + q)$  — one with no real root, i.e.  $p^2 < 4q$ , so it cannot be split into real linear factors — contributes  $\frac{Bx + C}{x^2 + px + q}$ ; a repeated one  $(x^2 + px + q)^m$  contributes one such term over each power up to  $m$ .

In every case the fragment's numerator sits one degree below its denominator — a constant over a linear factor, a linear  $Bx + C$  over a quadratic — and a factor raised to a power gets one fragment for each power up to that exponent, exactly the freedom needed to match an arbitrary proper numerator. The constants are then uniquely determined by  $P$  and  $Q$ .

That such a decomposition always exists, and with only one choice of constants, is a fact of linear algebra: matching numerators forces a system of linear equations in the unknowns that has exactly one solution. We take the *shape* as given and spend the effort where the work really is — pinning down the constants.

#### STRATEGY A.2 Finding the constants

1. If the fraction is *improper* (numerator degree  $\geq$  denominator degree), divide first:  $P/Q = (\text{a polynomial}) + (\text{proper remainder})/Q$ , and decompose only the remainder.
2. Factor the denominator completely and write the template from Proposition A.7 with the constants left as unknowns.
3. Multiply through by  $Q(x)$  to clear denominators, then solve for the constants — either by matching coefficients of like powers of  $x$ , or by substituting convenient values of  $x$  (a factor's own root kills every other term at once).

The substitution trick in step 3 is the cover-up shortcut: to get the constant over a distinct linear factor  $(x - a)$ , set  $x = a$  everywhere except in that one factor. It reaches only the top constant of a repeated factor, and an irreducible quadratic has no real root to substitute at all — those remaining constants come from matching coefficients.

**EXAMPLE A.4**

Decompose  $\frac{5x-1}{(x-1)(x+2)}$ .

**SOLUTION**

Two distinct linear factors call for the template  $\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$ . Multiplying through by  $(x-1)(x+2)$  gives

$$5x - 1 = A(x + 2) + B(x - 1).$$

Setting  $x = 1$  leaves  $4 = 3A$ , so  $A = \frac{4}{3}$ ; setting  $x = -2$  leaves  $-11 = -3B$ , so  $B = \frac{11}{3}$ . Hence

$$\frac{5x-1}{(x-1)(x+2)} = \frac{4/3}{x-1} + \frac{11/3}{x+2}.$$

Each substitution was a root of one factor, which zeroed the other term and handed back a single constant — the cover-up shortcut in action.

**EXAMPLE A.5**

Decompose each.

1.  $\frac{x}{(x-1)^2}$ , with a repeated linear factor.
2.  $\frac{x+1}{x(x^2+1)}$ , with an irreducible quadratic factor.

**SOLUTION**

1. The repeated factor  $(x-1)^2$  needs both powers:  $\frac{x}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$ . Clearing denominators,  $x = A(x-1) + B$ . Setting  $x = 1$  gives  $B = 1$ ; matching the coefficient of  $x$  gives  $A = 1$ . So  $\frac{x}{(x-1)^2} = \frac{1}{x-1} + \frac{1}{(x-1)^2}$ .
2. The quadratic  $x^2 + 1$  does not factor over the reals — there is no real root to cover up — so it carries a linear numerator, and its constants come from matching coefficients:  $\frac{x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$ . Clearing denominators,  $x+1 = A(x^2+1) + (Bx+C)x = (A+B)x^2 + Cx + A$ . Matching powers: the constant term gives  $A = 1$ , the coefficient of  $x$  gives  $C = 1$ , and the coefficient of  $x^2$  gives  $A+B = 0$ , so  $B = -1$ . Hence  $\frac{x+1}{x(x^2+1)} = \frac{1}{x} + \frac{1-x}{x^2+1}$ .

**CAUTION**

Partial fractions applies only to a *proper* fraction. Faced with  $\frac{x^2}{x-1}$ , divide first:  $x^2 = (x-1)(x+1)+1$ , so  $\frac{x^2}{x-1} = x+1 + \frac{1}{x-1}$ . No combination of the form  $\frac{A}{x-1}$  alone can reproduce a numerator that outgrows its denominator — the polynomial part must be divided out before the templates apply.

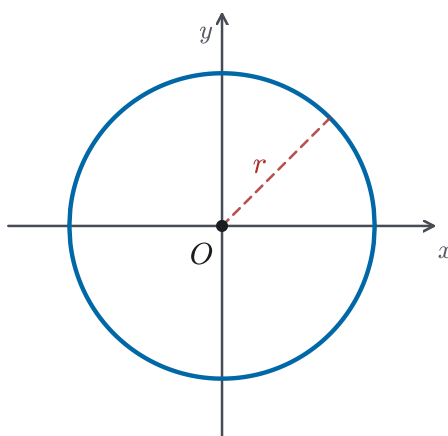
Every fragment this method produces —  $\frac{A}{x-a}$ ,  $\frac{A}{(x-a)^k}$ , and  $\frac{Bx+C}{(x^2+px+q)^k}$  — has a standard antiderivative, which is why partial fractions is the standard opening move for integrating a rational function. The algebra that gets you there is complete here.

## A.5 The Conic Sections

Slice a cone with a flat plane and the rim of the cut is a *conic section*: a circle, an ellipse, a parabola, or a hyperbola, according to the angle of the slice. Each also has a cleaner description in terms of distance, and each has a *standard equation* — the simplest form, obtained by placing the curve's centre or vertex at the origin. This tour records the four standard forms and the features to read straight off them.

**The circle.** A circle is the set of points a fixed distance  $r$  — the *radius* — from a centre; it is the conic already familiar from precalculus, and the template the other three vary. Centred at the origin, the condition  $\sqrt{x^2+y^2} = r$  squares to the standard equation

$$x^2 + y^2 = r^2.$$



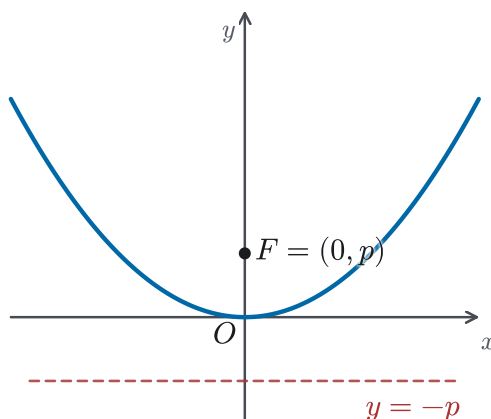
A circle: every point lies a fixed distance  $r$  from the centre  $O$ .

**Figure A.1** A circle of radius  $r$  centred at the origin:  $x^2 + y^2 = r^2$ .

**The parabola.** Fix a point  $F$  (the *focus*) and a line (the *directrix*). A parabola is the set of points equidistant from the two. Take focus  $F = (0, p)$  with  $p > 0$  and directrix  $y = -p$ . Equating a point's distance to the focus with its distance to the directrix — the latter being the vertical gap  $y + p$  down to the line  $y = -p$  — gives  $x^2 + (y - p)^2 = (y + p)^2$ , which simplifies to the standard equation

$$x^2 = 4py,$$

a curve opening upward from its vertex at the origin. (Putting the focus on the  $x$ -axis instead gives the sideways form  $y^2 = 4px$ .)



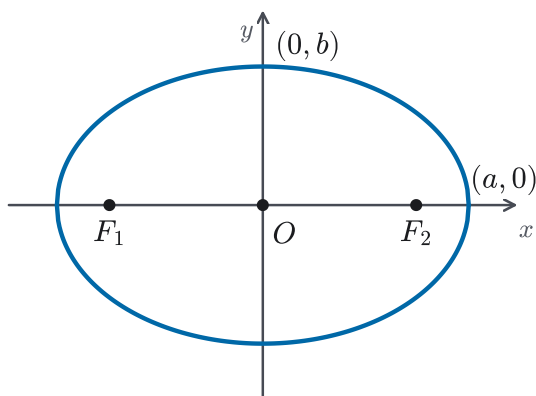
A parabola  $x^2 = 4py$ : every point is equidistant from the focus  $F$  and the directrix  $y = -p$ .

**Figure A.2** A parabola  $x^2 = 4py$ , with focus  $F = (0, p)$  and directrix  $y = -p$ .

**The ellipse.** Fix two points  $F_1, F_2$  (the *foci*). An ellipse is the set of points whose distances to the two foci add to a fixed total. Placing the foci at  $(\pm c, 0)$  and writing the total as  $2a$  leads to the standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad b^2 = a^2 - c^2,$$

with  $a > b > 0$ . The number  $b$  has a clean meaning: at the top of the ellipse,  $(0, b)$ , symmetry makes the two focal distances equal, and since they sum to  $2a$  each equals  $a$ ; the right triangle with legs  $b$  and  $c$  and hypotenuse  $a$  then gives  $a^2 = b^2 + c^2$  — the relation above. The curve crosses the axes at  $(\pm a, 0)$  and  $(0, \pm b)$ , and the longer semi-axis  $a$  runs along the line through the foci. Because the foci sit *inside* the curve,  $c < a$ , and  $c = \sqrt{a^2 - b^2}$  places them between the centre and the vertices.



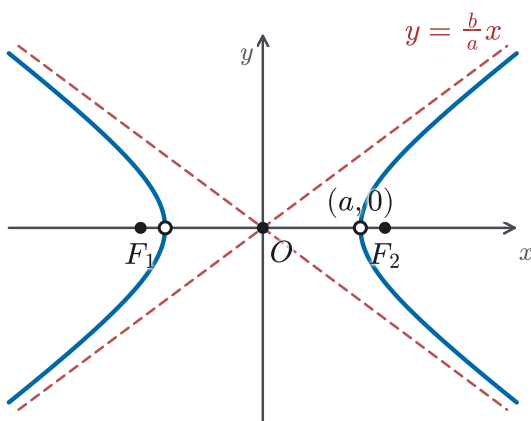
An ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  with  $a > b$ : the distances to the two foci  $F_1, F_2$  sum to a constant.

**Figure A.3** An ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  with foci  $F_1, F_2$  at  $(\pm c, 0)$ , where  $c = \sqrt{a^2 - b^2}$ .

**The hyperbola.** Keep two foci, but now ask for a fixed *difference* of distances rather than a sum. With foci at  $(\pm c, 0)$  and the difference written  $2a$ , the standard equation is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad b^2 = c^2 - a^2.$$

The curve has two branches with vertices at  $(\pm a, 0)$ , and far from the centre it draws arbitrarily close to the two *asymptotes*  $y = \pm \frac{b}{a}x$  — the crossed lines whose slopes  $\pm b/a$  set how widely the branches open. This time the foci lie *beyond* the vertices, so  $c > a$ , and the relation turns into  $c = \sqrt{a^2 + b^2}$ : a *plus* where the ellipse had a minus, precisely because the focus has moved from inside the curve to outside it.



A hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ : two branches with vertices at  $(\pm a, 0)$ , approaching the asymptotes  $y = \pm \frac{b}{a}x$ .

**Figure A.4** A hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ , with vertices  $(\pm a, 0)$ , foci  $(\pm c, 0)$ , and asymptotes  $y = \pm \frac{b}{a}x$ .

**EXAMPLE A.6**

Identify the curve  $4x^2 + 9y^2 = 36$  and locate its foci.

**SOLUTION**

Divide through by 36 to reach standard form:

$$\frac{x^2}{9} + \frac{y^2}{4} = 1.$$

Both squared terms are added, so the curve is an ellipse, with  $a^2 = 9$  and  $b^2 = 4$  — that is  $a = 3$  and  $b = 2$ . Since  $a > b$ , the major axis is horizontal, and  $c = \sqrt{a^2 - b^2} = \sqrt{9 - 4} = \sqrt{5}$ . The foci therefore sit at  $(\pm\sqrt{5}, 0)$ , about  $(\pm 2.24, 0)$ . Had the sign between the terms been a minus, the same reading would have named a hyperbola instead, with  $c = \sqrt{a^2 + b^2}$ .

Those are the four curves, their standard equations, and the features — centre, radius, focus, directrix, foci, axes, and asymptotes — that each equation puts on display. Moving a centre or vertex off the origin only replaces  $x$  and  $y$  with  $(x - h)$  and  $(y - k)$ ; the shapes, and everything this tour named, are unchanged.

Read backwards, that last remark is a tool. A second-degree equation handed over in *expanded* form —  $x^2 + y^2 - 6x + 4y = 12$ , say — hides its centre until you *complete the square* in each variable, grouping each pair and adding the square of half the linear coefficient to both sides:

$$\begin{aligned}(x^2 - 6x + 9) + (y^2 + 4y + 4) &= 12 + 9 + 4, \\ \text{that is } (x - 3)^2 + (y + 2)^2 &= 25,\end{aligned}$$

a circle of radius 5 centred at  $(3, -2)$ . The same single move,  $x^2 + px + q = (x + \frac{p}{2})^2 + (q - \frac{p^2}{4})$ , is what later lets an integral such as  $\int \frac{dx}{x^2 + px + q}$  be recentred and evaluated (Chapter 8). Completing the square is the one algebraic step that turns such an expression — one with no  $xy$  cross-term — into a standard one; a genuine  $xy$  term would call for a rotation of the axes as well, a step beyond this tour.

## A.6 Absolute Value and Inequalities

The absolute value  $|x|$  is the size of  $x$  with its sign discarded —  $|3| = 3$  and  $|-3| = 3$ . Its real use in calculus is geometric:  $|x - a|$  is the *distance* between  $x$  and  $a$  on the number line, and every  $\varepsilon$ - $\delta$  statement is a sentence about such distances. Reading absolute-value inequalities fluently, in both directions, is therefore a standing prerequisite rather than a one-off trick.

The single most useful reading turns a distance bound into an interval. For  $\delta > 0$ ,

$$|x - a| < \delta \iff a - \delta < x < a + \delta,$$

since “within  $\delta$  of  $a$ ” is exactly “no further than  $\delta$  to either side.” The set on the right is the  $\delta$ -*neighbourhood* of  $a$ : an open interval of width  $2\delta$  centred at  $a$ .

Closing the inequality closes the interval —  $|x - a| \leq \delta \Leftrightarrow a - \delta \leq x \leq a + \delta$  — and reversing it turns the region inside out:  $|x - a| > \delta$  describes the two rays *outside*,  $x < a - \delta$  or  $x > a + \delta$ . This is the unpacking behind every limit and continuity statement, and behind the interval of convergence of a power series, where  $|x - a| < R$  reads as “ $x$  lies within  $R$  of the centre  $a$ .”

**EXAMPLE A.7**

Solve  $|2x - 3| < 5$ , and describe the solution as a neighbourhood.

**SOLUTION**

Unpack the bound into a double inequality and work the two sides together:

$$\begin{aligned} |2x - 3| < 5 &\Leftrightarrow -5 < 2x - 3 < 5 \\ &\Leftrightarrow -2 < 2x < 8 \\ &\Leftrightarrow -1 < x < 4. \end{aligned}$$

The solution is the open interval  $(-1, 4)$ . Its midpoint is  $\frac{-1+4}{2} = \frac{3}{2}$  and its half-width is  $\frac{5}{2}$ , so the same set reads  $|x - \frac{3}{2}| < \frac{5}{2}$  — a neighbourhood of  $\frac{3}{2}$ . Dividing the original bound by 2 shows why:  $|2x - 3| = 2|x - \frac{3}{2}|$ , so  $|2x - 3| < 5$  is the same demand as  $|x - \frac{3}{2}| < \frac{5}{2}$ .

One inequality outranks all others for the proofs in this book. It bounds the size of a sum by the sizes of its parts, and it is the lever behind almost every  $\varepsilon$ - $\delta$  estimate.

**PROPOSITION A.8 Triangle inequality**

For all real numbers  $x$  and  $y$ ,

$$|x + y| \leq |x| + |y|,$$

and consequently  $||x| - |y|| \leq |x - y|$ .

**PROOF**

Every real number lies between the negative and the positive of its own size:  $-|x| \leq x \leq |x|$ , and likewise  $-|y| \leq y \leq |y|$ . Adding the two chains term by term,

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

which says exactly that  $x + y$  is no further than  $|x| + |y|$  from zero — that is,  $|x + y| \leq |x| + |y|$ . For the second form, apply the first to  $|x| = |(x - y) + y| \leq |x - y| + |y|$ , so that  $|x| - |y| \leq |x - y|$ ; swapping  $x$  and  $y$  gives  $|y| - |x| \leq |x - y|$  as well, and the two together are precisely  $||x| - |y|| \leq |x - y|$ .  $\square$

In practice the triangle inequality appears whenever an error must be split and controlled piece by piece: to force  $|f(x) - L|$  below  $\varepsilon$ , bound it by a sum of quantities each held under  $\frac{\varepsilon}{2}$ , and the total stays under  $\varepsilon$ . The chapters that prove limits, continuity, and convergence lean on this move constantly.



**CAUTION**

An absolute-value inequality is *not* disposed of by simply dropping the bars. The statement  $|x - a| < \delta$  unpacks to the *pair*  $-\delta < x - a < \delta$ , not to  $x - a < \delta$  alone — the lower bound is half of what was said. Keeping only the upper half retains the points on one side of  $a$  and silently discards the rest of the neighbourhood.